Chapter 2
Text processing

Abstract Text processing and document creation are an important part of the open-source world. Most open-source tools come with documentation which is written using SGML or TeXinfo (both described in this chapter). Complete tools such as OpenOffice, devote significant portion to word processing. In this chapter we describe open-source tools for text document processing. We discuss OpenOffice, TEXand LATEX, SGML, and TeXinfo. Page layout tool, Scribus, and document classification/citation tools are described.

Contents

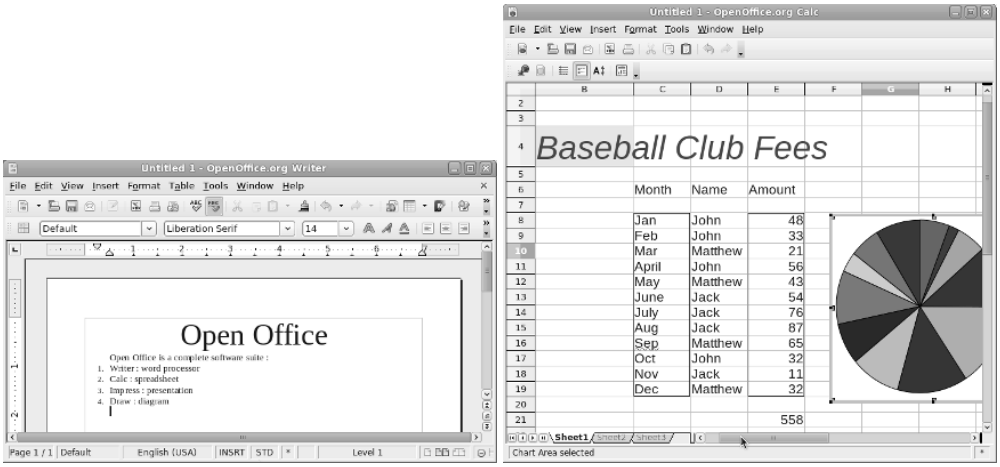
2.1 OpenOffice.org Suite **35**
2.2 TeX and LaTeX **37**
2.3 Scribus **41**
2.4 Document classification software **42**
2.5 Wiki **43**
2.6 Conclusion **43**

Even with all the advent in graphical user interfaces, electronic document processing and production remains an important part of any product development project. There are many open source document preparation and markup systems; we discuss some of them in this chapter.

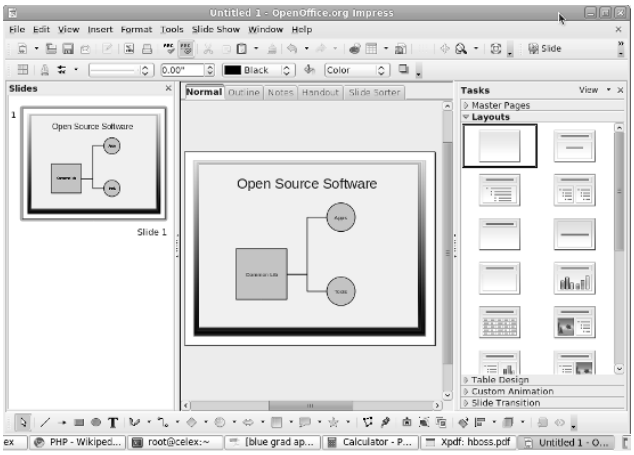
2.1 OpenOffice.org Suite

The OpenOffice.org system is an open-source office software suite which includes (i) word processing, (ii) spreadsheets, (iii) presentation, (iv) graphics and diagrams, and (v) database. It is available in many languages and on many platforms. It can read-write data files from many different software packages. Example of OpenOffice.org tools is shown in Figure 2.1.

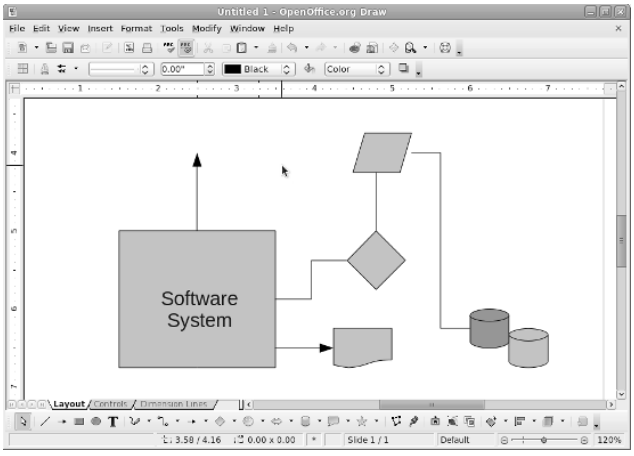
S. Koranne, *Handbook of Open Source Tools*,
DOI 10.1007/978-1-4419-7719-9_2, © Springer Science+Business Media, LLC 2011



(a) Writer (word processor) and Calc (spreadsheet)



(b) Impress (presentation software)



(c) OpenOffice Draw

Fig. 2.1 OpenOffice.org System

2.2 TeX and LaTeX

Consider the example of an article written in LaTeX:

```
%% Simple document in LaTeX
\documentclass{article}[12pt]
\usepackage{a4wide}
\usepackage{amsmath}
\usepackage{times}
\title{Introduction to Writing with \LaTeX}
\author{Sandeep Koranne}
\begin{document}
\maketitle
\begin{abstract}
This short document demonstrates the use of \LaTeX in writing
professional quality typeset articles. In
Section~\ref{section:math} we give an example
of using \TeX for typesetting mathematics, which was the
reason for its invention. We conclude in
Section~\ref{section:conclusion}.
\end{abstract}

\section{Introduction}

\section{Typesetting Mathematics}
\label{section:math}
Consider the following quadratic equation in $x$:
\begin{equation}
ax^2 + bx + c = 0 \label{eqn:quadratic}
\end{equation}
Equation~\ref{eqn:quadratic} has exactly two roots
$x = \frac{-b \pm \sqrt{ b^2 - 4ac}}{2a}$. The same
solution can be written as:
\begin{eqnarray}
x_1 & = & \frac{-b + \sqrt{ b^2 - 4ac}}{2a} \\
x_2 & = & \frac{-b - \sqrt{ b^2 - 4ac}}{2a}
\end{eqnarray}

\section{Conclusion}
\label{section:conclusion}
The use of references within the article is made
easier by using labels and references are automatically
generated.
\end{document}
```

We process this article with LaTeXto generate a device independent file (DVI) as shown in Figure 2.2.

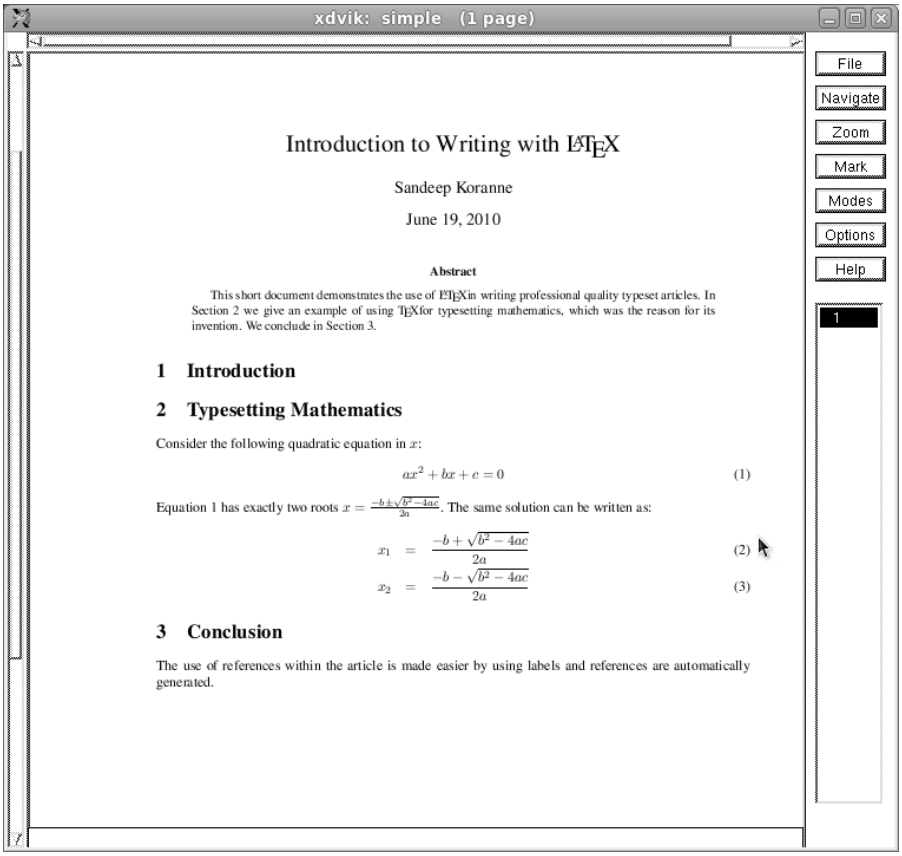


Fig. 2.2 Example of article written in LaTeX

2.2.1 Lout Typesetting System

Lout is a typesetting software similar in concept to TeXbut implemented quite differently. It has a markup language, and can generate PostScript from Lout documents.

2.2.2 SGML Processing

Standard Generalized Markup Language. Most GNU manuals are written using markup language as opposed to WYSIWYG word processors. Markup language is identified by the following:

- structure as opposed to formatting: the markup should describe the structure of the material as opposed to any specific formatting to be applied to it at any given time,
- automatic processing: the structure imposed on the source document should be sufficient to allow automated processing of data using software tools. We discuss such an automated document generator for source code in Section 3.10.

As a documentation system SGML was designed to enable the sharing of machine readable large corpus of documents from various agencies of the government, law, and business applications. SGML is mostly associated with textual data, and XML and HTML are closely related to SGML.

2.2.3 Texinfo : GNU Documentation System

Texinfo is the official documentation format of the GNU project. It is based on the concept of markup language, and from a single source document many different formats, including 'dvi', 'html', 'info', and 'pdf' can be automatically generated. An example of a '.texi' file from GCC is shown below:

```
@node Compatibility
@chapter Binary Compatibility
@cindex binary compatibility
@cindex ABI
@cindex application binary interface

Binary compatibility encompasses several related concepts:

@table @dfn
@item application binary interface (ABI)
The set of runtime conventions followed by all of
the tools that deal with binary representations of a program
```

Texinfo files can be converted to output formats using the `makeinfo` program. The output format can be specified using the following options:

```
--docbook    : docbook in XML
--html       : output in HTML
--xml        : Texinfo XML
--plaintext  : plain text
```

The related tool `troff` generated `man` pages from descriptions formatted in a similar manner.

2.2.4 LyX Frontend

LyX is a word-processor front end to the TEXand LATEXtypesetting systems. Modeled after the WYSIWYG (what you see is what you get) principle, authors who are not familiar with TEXcan use the GUI and commands of LyX to produce high quality manuscripts. An example of LyX in action is shown in Figure 2.3.

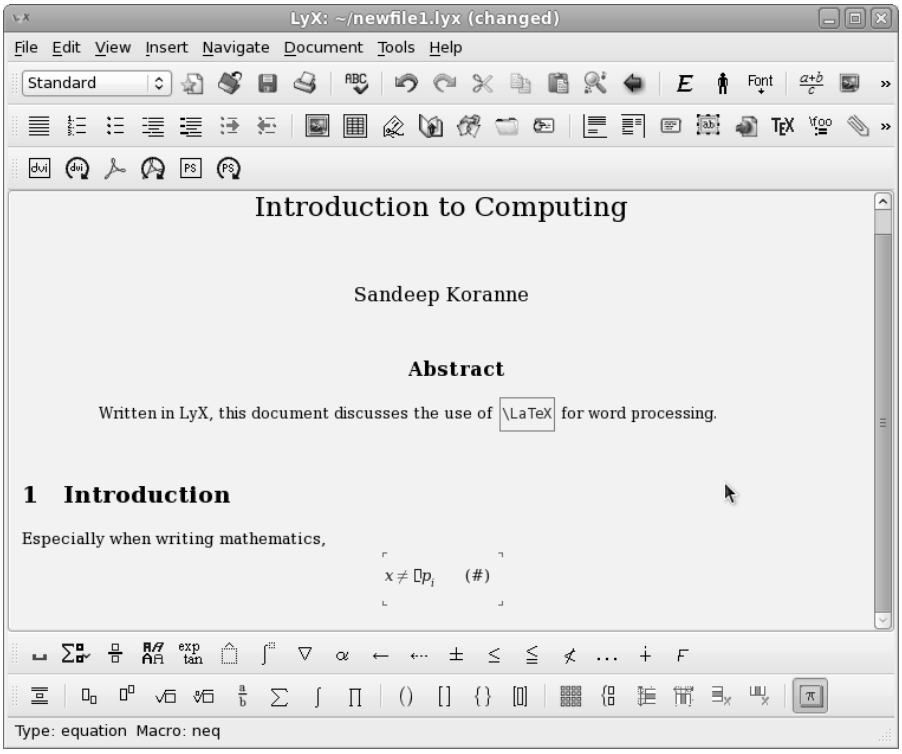


Fig. 2.3 LyX word processing software using LaTeX

2.2.5 Texmaker LaTeX Editor

Another frontend to LaTeXis TexMaker; but unlike LyX TexMaker is not a WYSIWYG word processor. The author using TexMaker has to be more familiar with LaTeX, although like a good GUI, TexMaker does provide menus and toolbars to automate and simplify many tasks. An example of TexMaker editing a chapter of this book is shown in Figure 2.4.

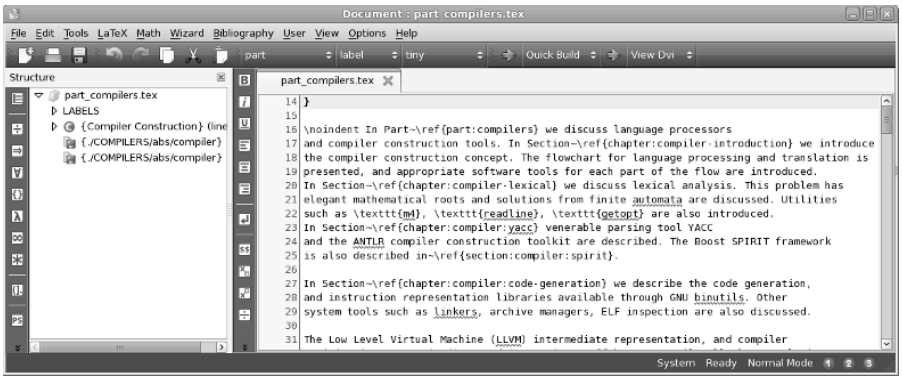


Fig. 2.4 TexMaker LaTeXfrontend

2.2.6 PostScript and PDF Support

The output of LaTeXis a DVI file, but more and more, PDF (portable document format) and PostScript output is also expected from the tools. There are a number of open-source programs which can generate and process PostScript and PDF files. Document viewers such as (i) ghostscript, (ii) evince, and (iii) xpdf are available. Ghostscript in particular has PostScript processing tools which can manipulate PostScript data for efficient printing, double-sided printing, and even format conversion from PostScript to PDF.

In a similar vein, the text to PostScript converter program `a2ps` can be used to convert TEXT files (including source code listing) to neatly formatted PostScript data.

2.3 Scribus

Unlike Lyx and TexMaker, Scribus is a *desktop publication* (DTP) software. Such software is often used when designing and printing brochures, pamphlets, menus, and other promotional material. An example of Scribus in designing a library information brochure is shown in Figure 2.5.

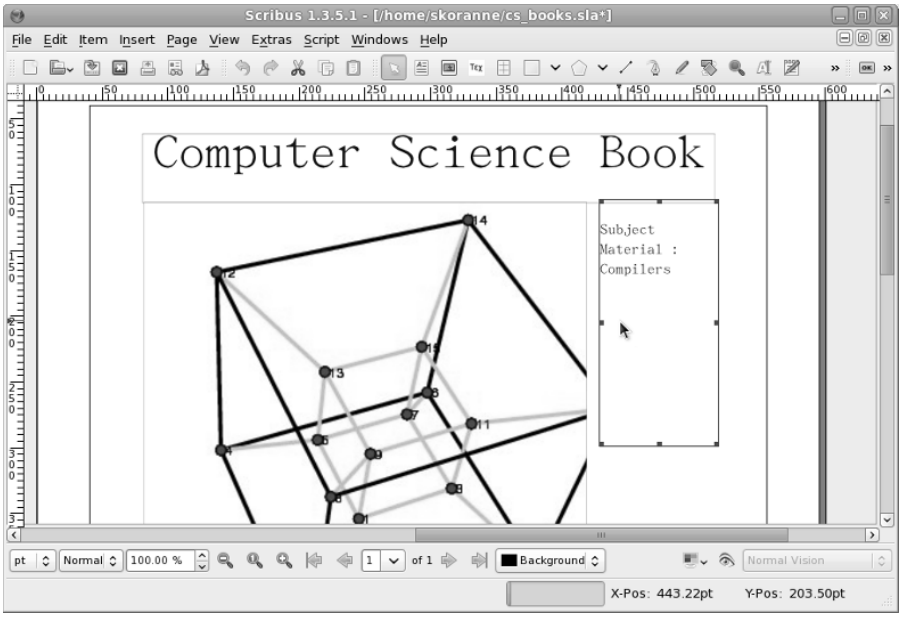


Fig. 2.5 Scribus desktop publishing program

2.3.1 Citation management

If using TEXor LATEX, the appropriate text citation tool is BibTex, which accepts an input '.bib' file containing citations of the form:

```
@ARTICLE{karger,
    author = {David R. Karger and Clifford Stein},
    title = {A new approach to the minimum cut problem},
    journal = {Journal of the ACM},
    year = {1996},  volume = {43}, pages = {601--640}
}
```

This article can be cited in a LATEXsource as:

```
In his paper Karger~\cite{karger} presented a
```

which produces output such as "paper Karger [18] says", where the number 18 is automatically calculated based on the citation order. The software also produces the list of used references for inclusion in the main text.

2.4 Document classification software

In addition to text document creation tools, engineers and scientists also need document classification, citation, and search/index tools. We discuss document file searching tools on GNU/Linux systems. We have already seen the GNU `find` tool for searching for file names in a hierarchical path. Since `find` actually traverses the file system its performance can be slow for large file systems.

2.4.1 GNU `locate`

GNU tools `locate` and its security enhanced version `slocate` provide an effective command line file searching tool for GNU/Linux systems. The command line for `locate` is:

```
$locate [options] file-pattern
```

GNU `locate` uses database prepared by `updatedb`, and thus is not bottlenecked by the file system access speed or size, and is significantly faster than using `find` when searching for files. By default `locate` does not check whether the file reported still exists in the system or not; moreover, `locate` reports on files which were present when the last database update was performed. Like many other UNIX tools, `locate` exits with status 0 if a match was found, 1 otherwise. The options accepted by `locate` are:

1. `-b`: match basename only (instead of wholename),

2. `-c`: print count of matching files (instead of names),
3. `-e`: report only files which still exist,
4. `-L`: follow symbolic links,
5. `-l <n>`: limit search to the first *n* matches only,
6. `-S`: print statistics about the database,
7. `-r`: use regex (regular expression) pattern matching.

The default database is located in `/var/lib/mlocate/mlocate.db`, running `locate -S` returns:

```
Database /var/lib/mlocate/mlocate.db:
65,147 directories
927,777 files
59,636,635 bytes in file names
20,767,916 bytes used to store database
```

A security enhanced version of `locate` also exists, and is called `slocate`. It only allows searches for files which have appropriate permission for the user who is performing the search.

2.5 Wiki

A Wiki is described as a collaborative software which is implemented as a website. A wiki allows multiple users to edit any page or to create new pages which can be used to maintain collaborative information between groups of users. An example of Wiki is Wikipedia, which is an encyclopedia maintained by a group of users. Wiki updates are written with markup languages using web browser. A starting wiki web page simply contains an Edit as well as a Save button. New pages in the wiki set can be created by the users and added as a link to existing pages. The singular point of the Wiki is that it allows any user to edit any page of the wiki website (of course, this feature can be selectively switched off). This allows for a Wiki to grow as more and more users add to the collective knowledge. Many wikis are completely open to the general public, while others may have certain restriction to allow rudimentary quality control on the content. Another very recent open-source addition is the Eureka Streams open social networking project.

2.6 Conclusion

Information dissemination is an integral part of scientific research and information science also benefits from the generation of high-quality documents. The software presented in this chapter can help the scientist present the research in a standardized and attractive manner. Above, we discussed text processing and document creation tools including OpenOffice, Wiki, Scribus, SGML, LaTeXand its various front-ends

(including LyX and Texmaker). Document management and searching tools, along-with citation tools were presented with examples.

Part II
Software Engineering and Libraries